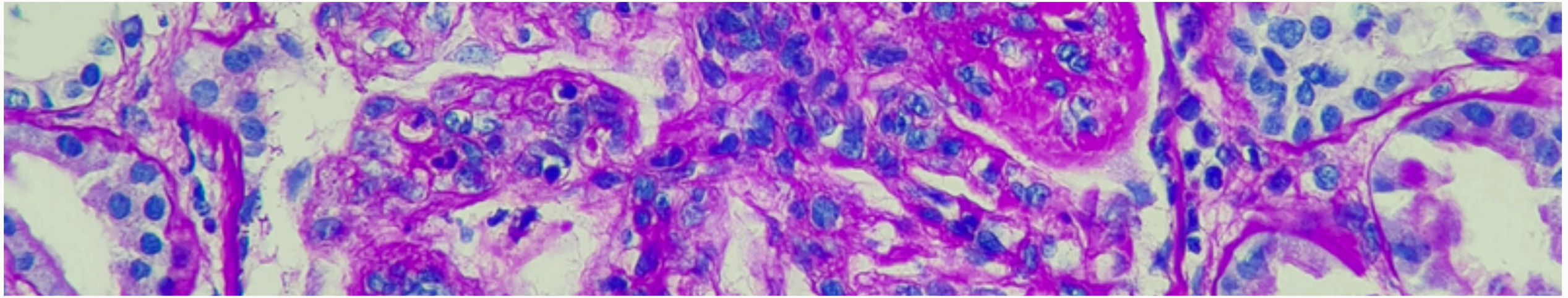
UNIVERSITY OF TORONTO MISSISSAGA

國立陽明交通大學 NATIONAL YANG MING CHIAO TUNG UNIVERSITY

Ko Lab

AUGUST 5, 2026



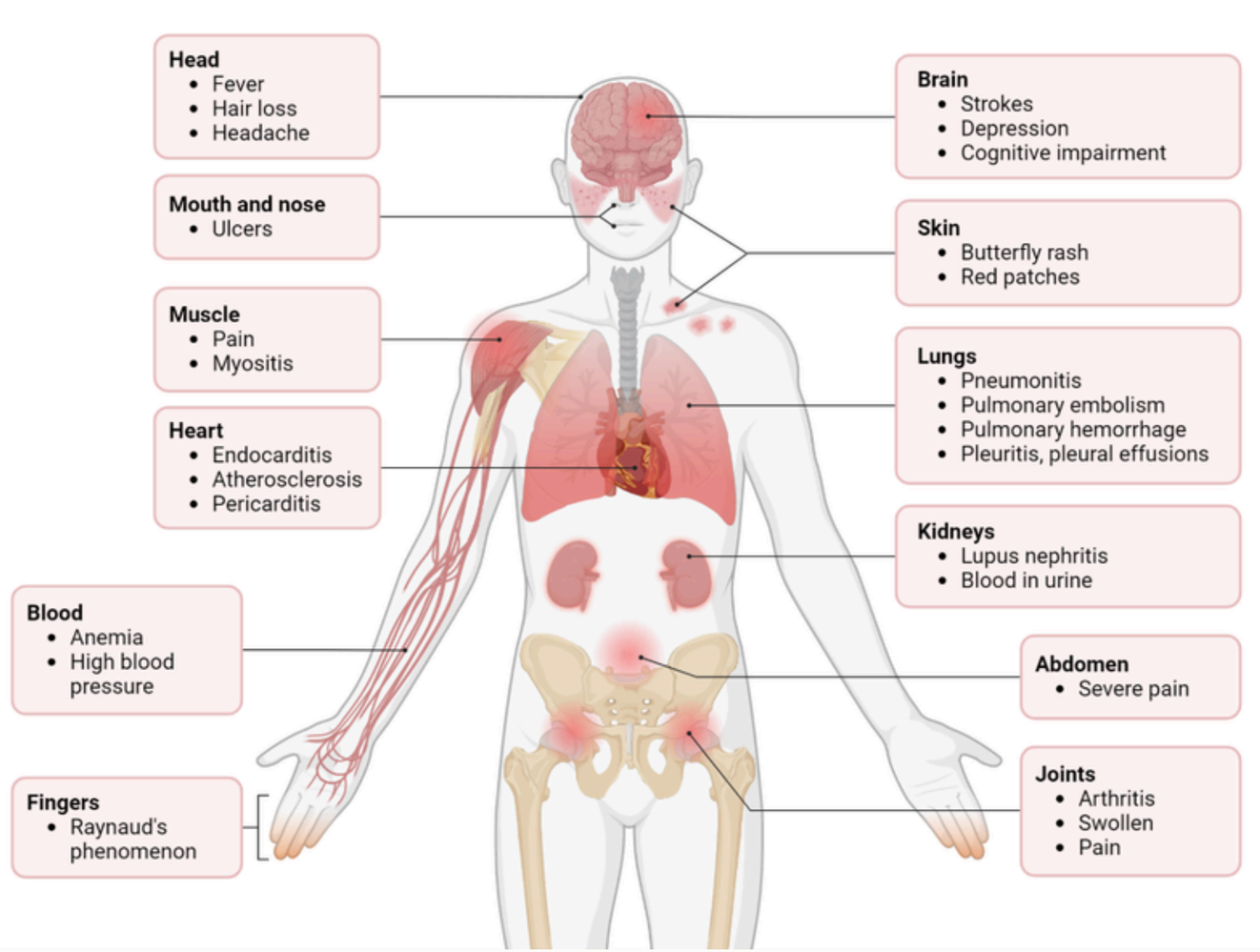
Lupus nephritis, shown above, can be a complication of systemic lupus. (Isabella Cueto, July 10, 2024).

IDENTIFYING THE DOMINANT B-CELL TRANSCRIPTIONAL SIGNATURES IN SYSTEMIC LUPUS ERYTHEMATOSUS AND LUPUS NEPHRITIS

Presented by: Sarah Olsen

PI: Professor Tai Ming Ko

Systemic Lupus Erythematosus (SLE)



Common symptoms of SLE. (Wendy Jiang, BioRender)

Chronic autoimmune disease

- Immune system attacks healthy tissues
- Multi-organ inflammation

Highly Heterogenous

- Different clinical responses to the same treatment

Systemic Lupus Erythematosus (SLE)



Common symptoms of SLE. (Wendy Jiang, BioRender)

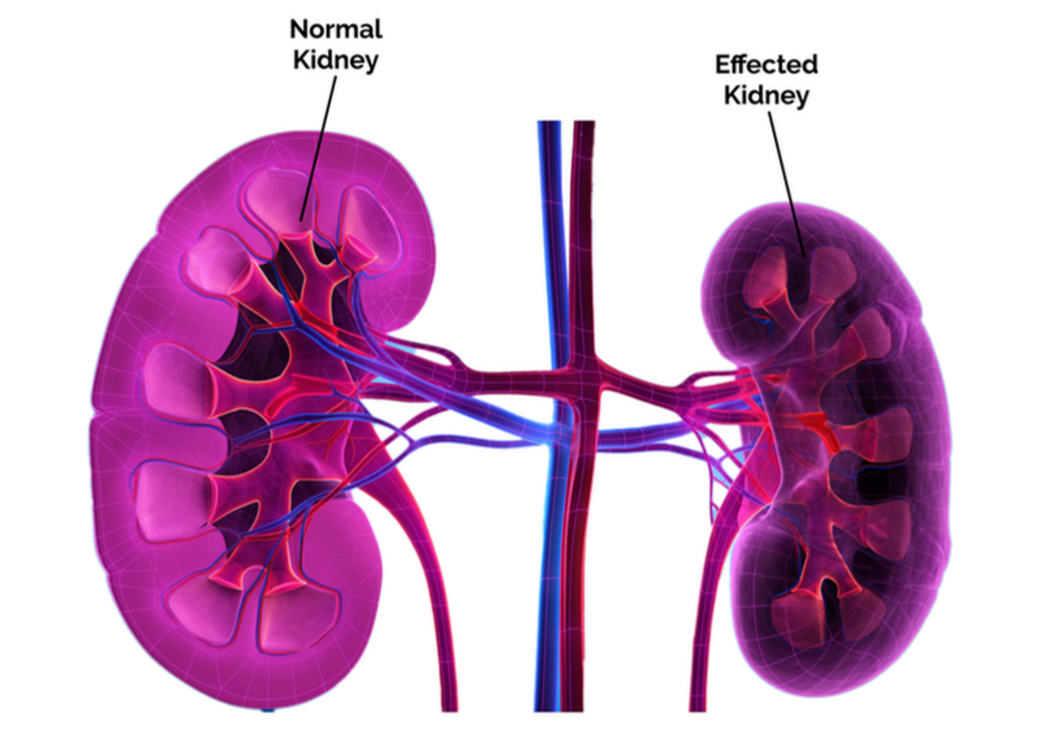
Chronic autoimmune disease

- Immune system attacks healthy tissues
- Multi-organ inflammation

Highly Heterogenous

- Different clinical responses to the same treatment

SLE can lead to LN...





(Treatians, Lupus Nephritis)

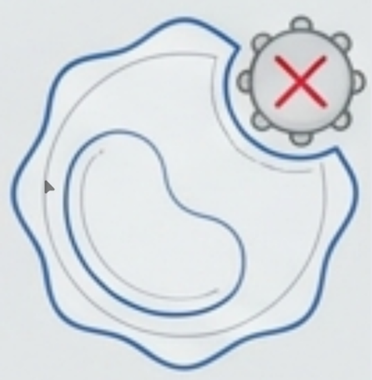
Lupus Nephritis (LN)

- Occurs in ~40–60% of SLE patients
- Major cause of morbidity and mortality
- Not all SLE patients develop LN

What drives progression from SLE to Lupus Nephritis?

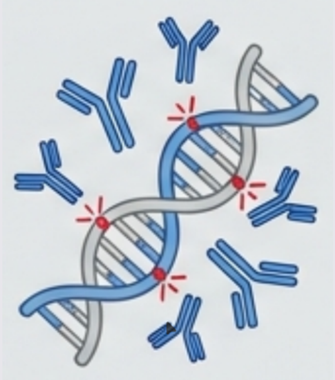
1

Loss of Self-Tolerance



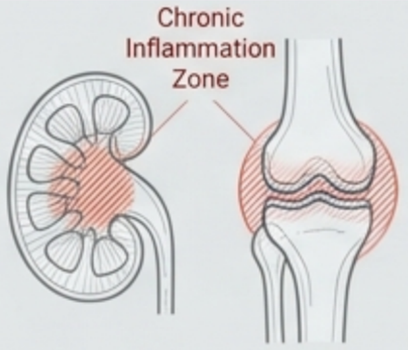
2

Autoantibodies



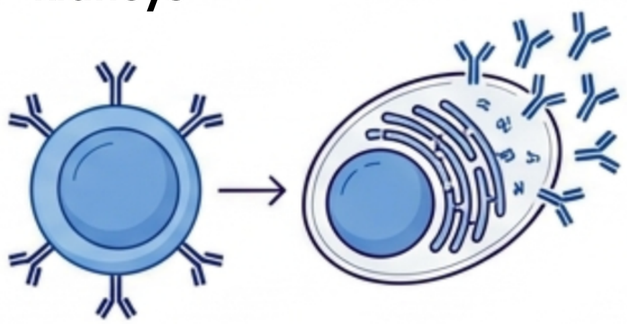
3

Tissue Damage



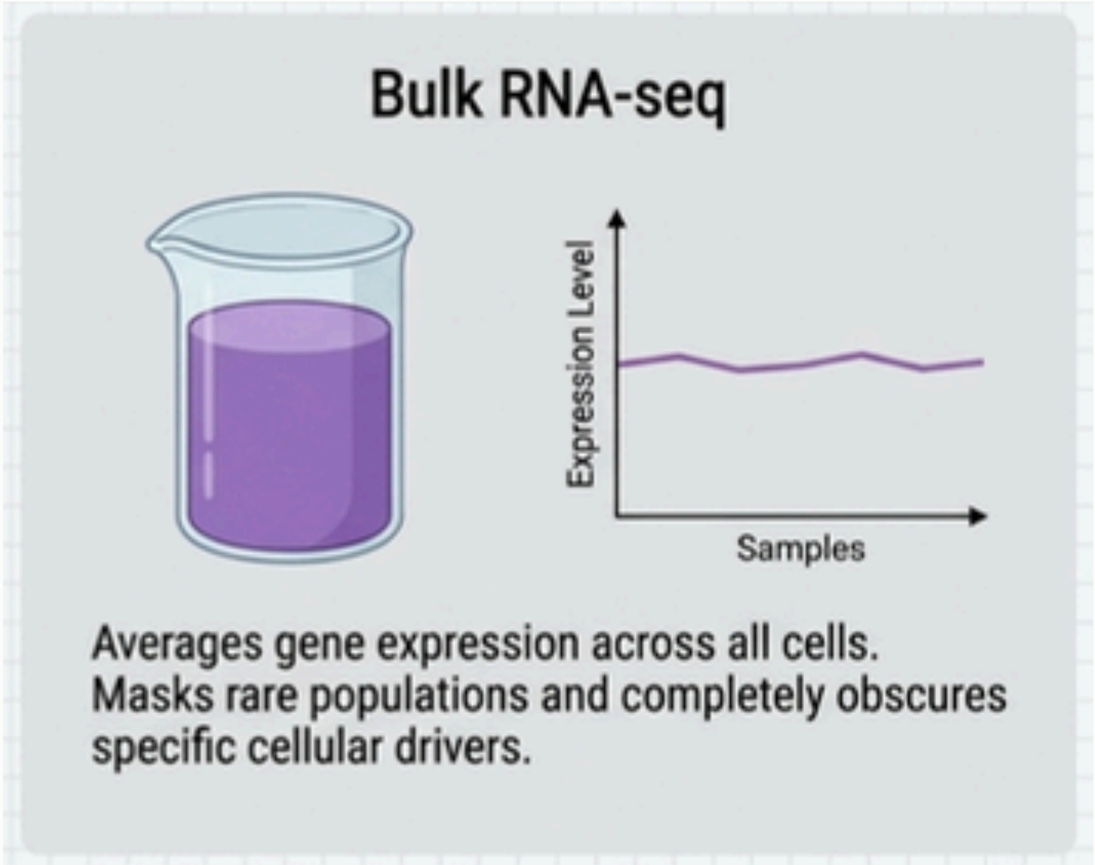
The B-Cell Role

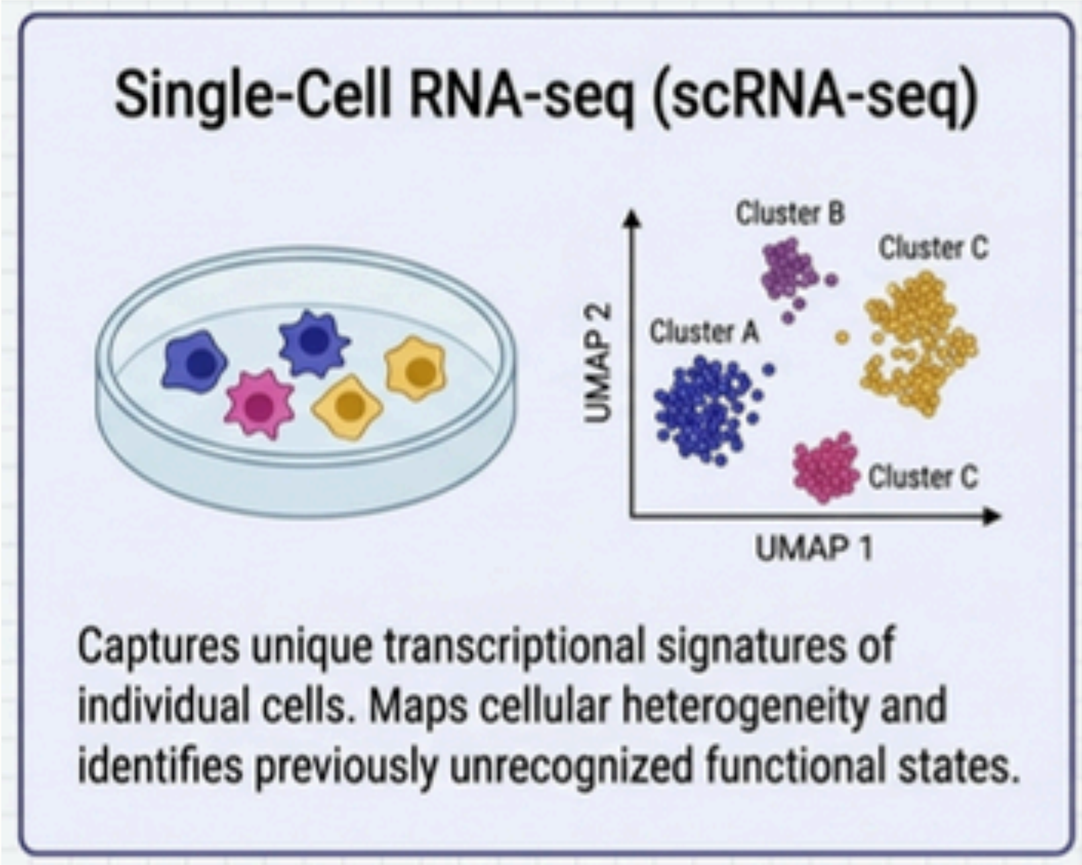
Dysregulated B-cell activity promotes chronic inflammation and immune complex deposition in the kidneys

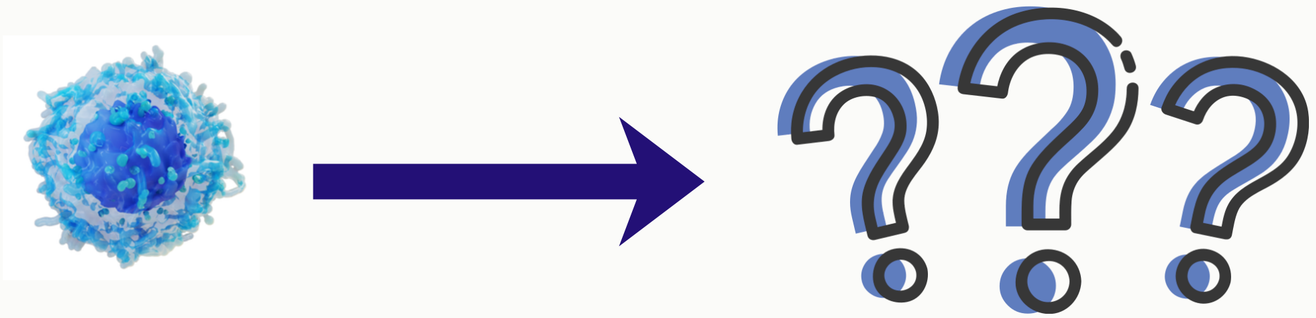


Resolving Cellular Heterogeneity

Single-cell resolution can provide deeper insight into B-cell subsets which drive inflammatory signals

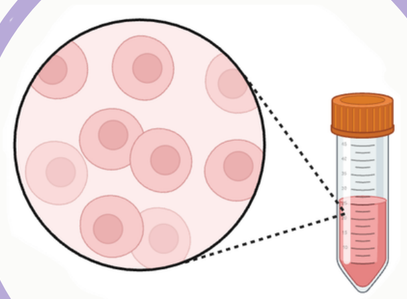






How does B-cell gene expression change during disease progression from SLE to LN?

Objective: Leveraging Single Cell RNA Sequencing to identify B-cell transcriptional programs that distinguish SLE from lupus nephritis and provide insight into disease progression.



PBMC samples
(3 HC, 3 SLE , 3 LN)

Illumina NovaSeq 6000

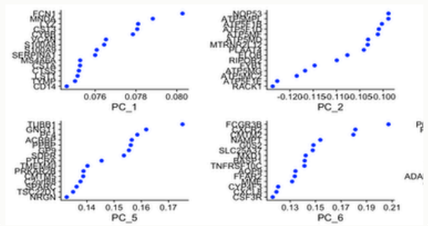


Single Cell RNA Sequencing



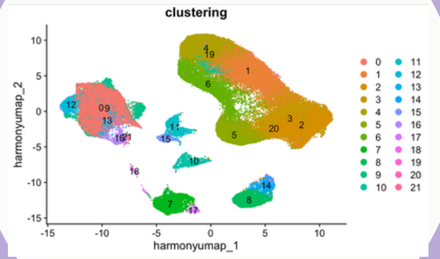
Quality Control, Normalization & Harmony Batch Correction

PCA



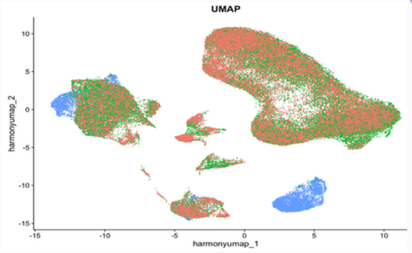
Reduce Data Dimensionality

Cell Custering



Identify Groups of Transcriptionally Similar Cells

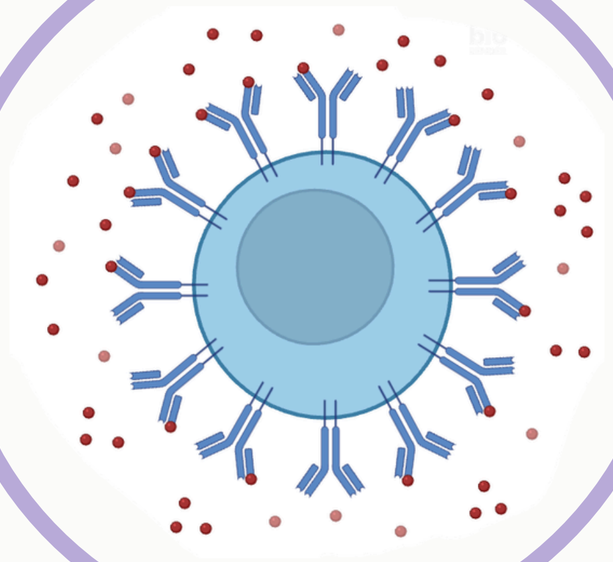
UMAP



2D Relationship Visualization

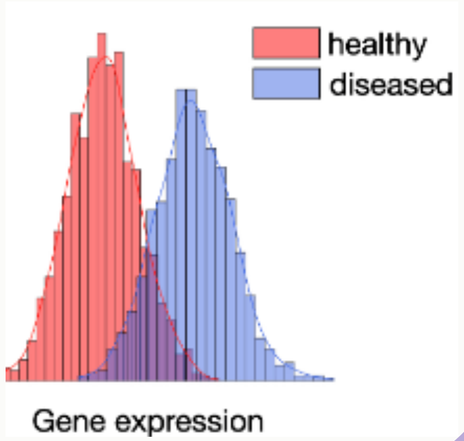
- B cells, naive
- Monocytes, CD14+
- Monocytes, CD16+
- NK cells
- T cells, CD4+, memory TREG
- T cells, CD4+, naive
- T cells, CD4+, naive TREG
- T cells, CD4+, naive, stimulated
- T cells, CD4+, TFH
- T cells, CD4+, Th1
- T cells, CD4+, Th1_17
- T cells, CD4+, Th17
- T cells, CD4+, Th2
- T cells, CD8+, naive
- T cells, CD8+, naive, stimulated

Cell Type Annotation

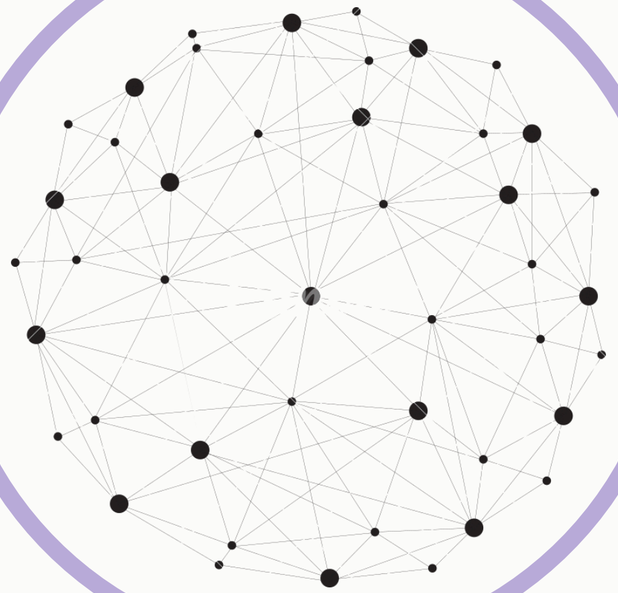


B-Cell Subsetting

DEG



Identify abnormal genes and pathways

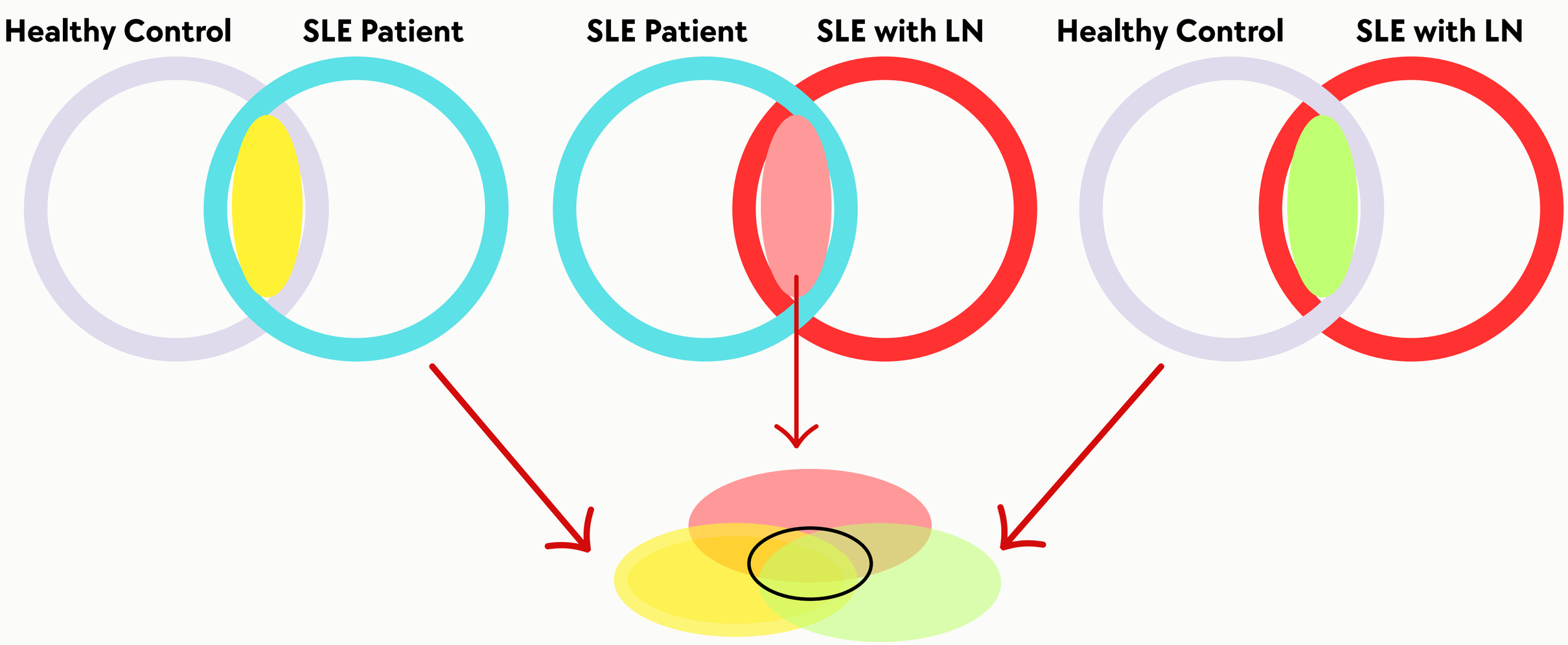


Pathway Enrichment Analysis

Study Design

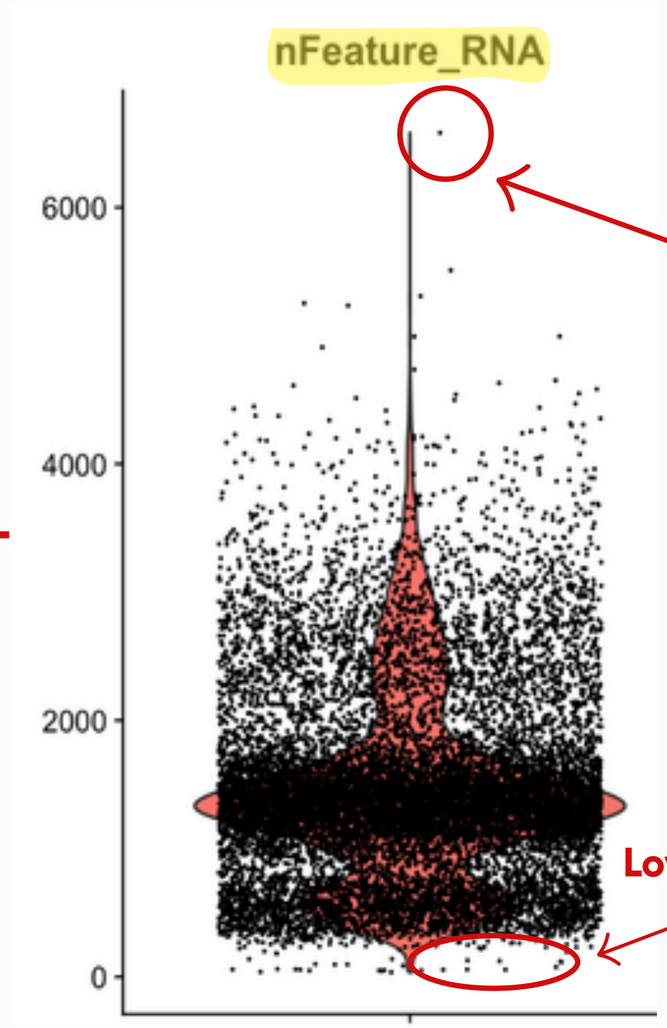


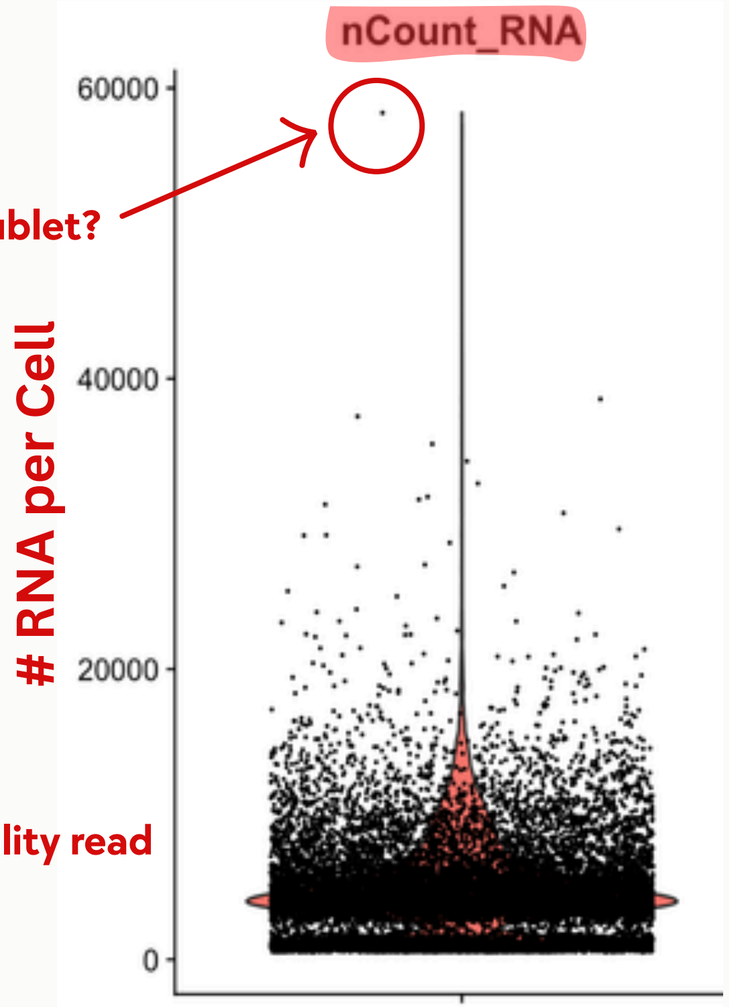
Healthy Control	SLE Patient	SLE Patient with LN
1	1	1
2	2	2
3	3	3

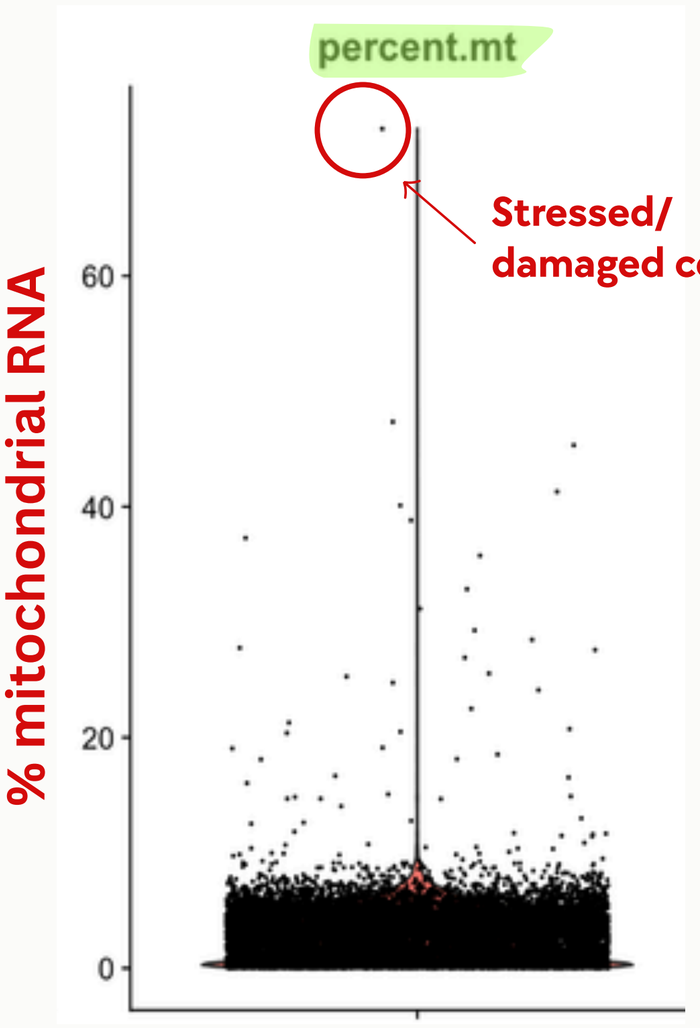

Healthy Control
SLE Patient
SLE Patient
SLE with LN
Healthy Control
SLE with LN
Shared DEGs

Quality Control









Detect low-quality cells and doublets

Measure sequencing depth

Exclude stressed or dying cells

```
#Filtered
```{r, warning=FALSE}
#label data
pre1 <- subset(pre1, subset = (nCount_RNA >= 1500)  & (nFeature_RNA >= 500) & (nFeature_RNA <= 3200) & (percent.mt < 20))
LN1 <- subset(LN1, subset = (nCount_RNA >= 1500)  & (nFeature_RNA >= 500) & (nFeature_RNA <= 3200) & (percent.mt < 20))
HC1 <- subset(HC1, subset = (nCount_RNA >= 1500)  & (nFeature_RNA >= 500) & (nFeature_RNA <= 3200) & (percent.mt < 20))

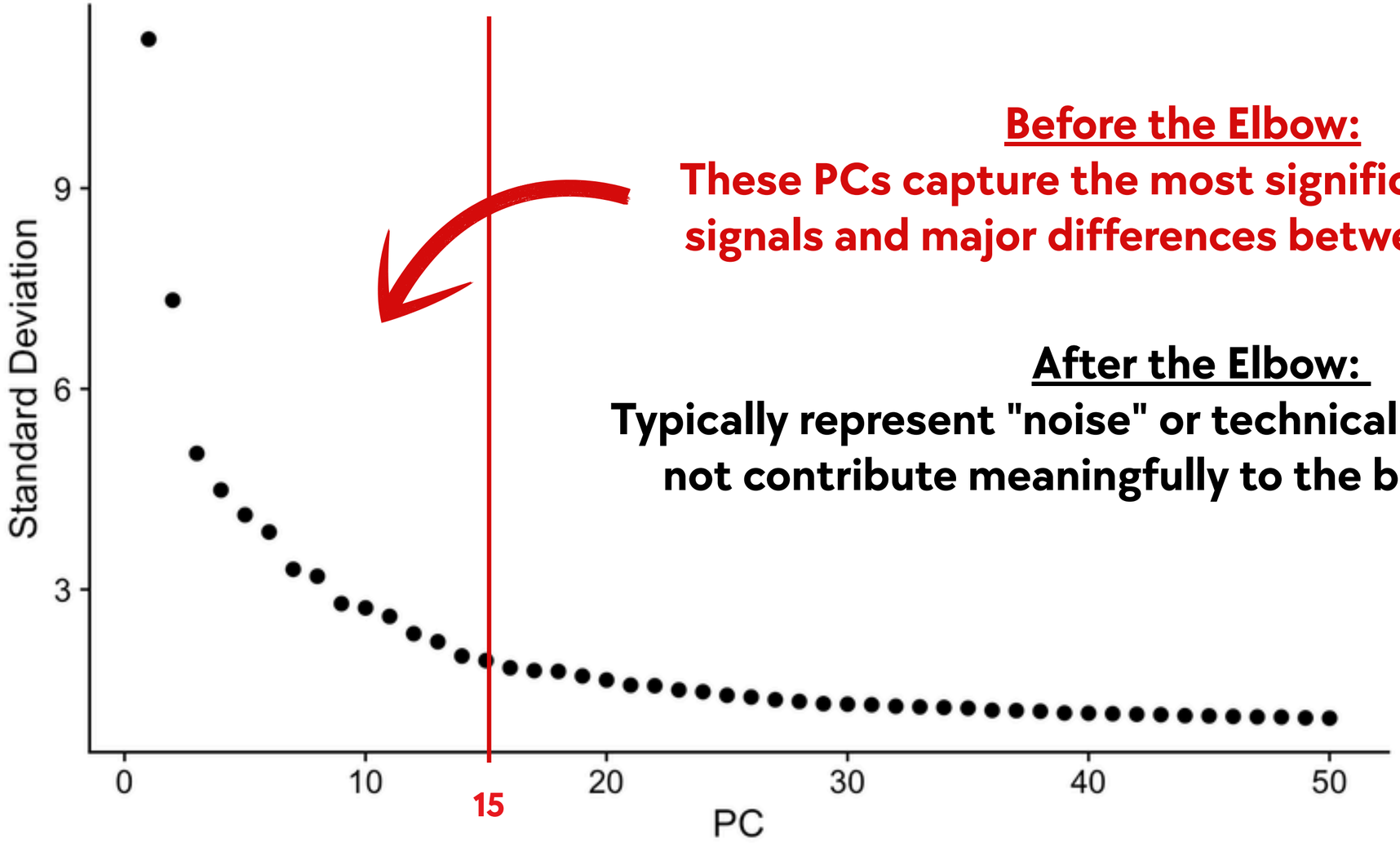

pre5 <- subset(pre5, subset = (nCount_RNA >= 1500)  & (nFeature_RNA >= 500) & (nFeature_RNA <= 3200) & (percent.mt < 20))
LN2 <- subset(LN2, subset = (nCount_RNA >= 1500)  & (nFeature_RNA >= 500) & (nFeature_RNA <= 3200) & (percent.mt < 20))
HC2 <- subset(HC2, subset = (nCount_RNA >= 1500)  & (nFeature_RNA >= 500) & (nFeature_RNA <= 3200) & (percent.mt < 20))


pre6 <- subset(pre6, subset = (nCount_RNA >= 1500)  & (nFeature_RNA >= 500) & (nFeature_RNA <= 3200) & (percent.mt < 20))
LN3 <- subset(LN3, subset = (nCount_RNA >= 1500)  & (nFeature_RNA >= 500) & (nFeature_RNA <= 3200) & (percent.mt < 20))
HC3 <- subset(HC3, subset = (nCount_RNA >= 1500)  & (nFeature_RNA >= 500) & (nFeature_RNA <= 3200) & (percent.mt < 20))
```

Low Total RNA Counts **Low Gene Complexity** **Unusually High Gene Detection** **HIgh mtRNA Content**

Removes Cells With:
```

Principle Component Analysis (PCA)



Before the Elbow:

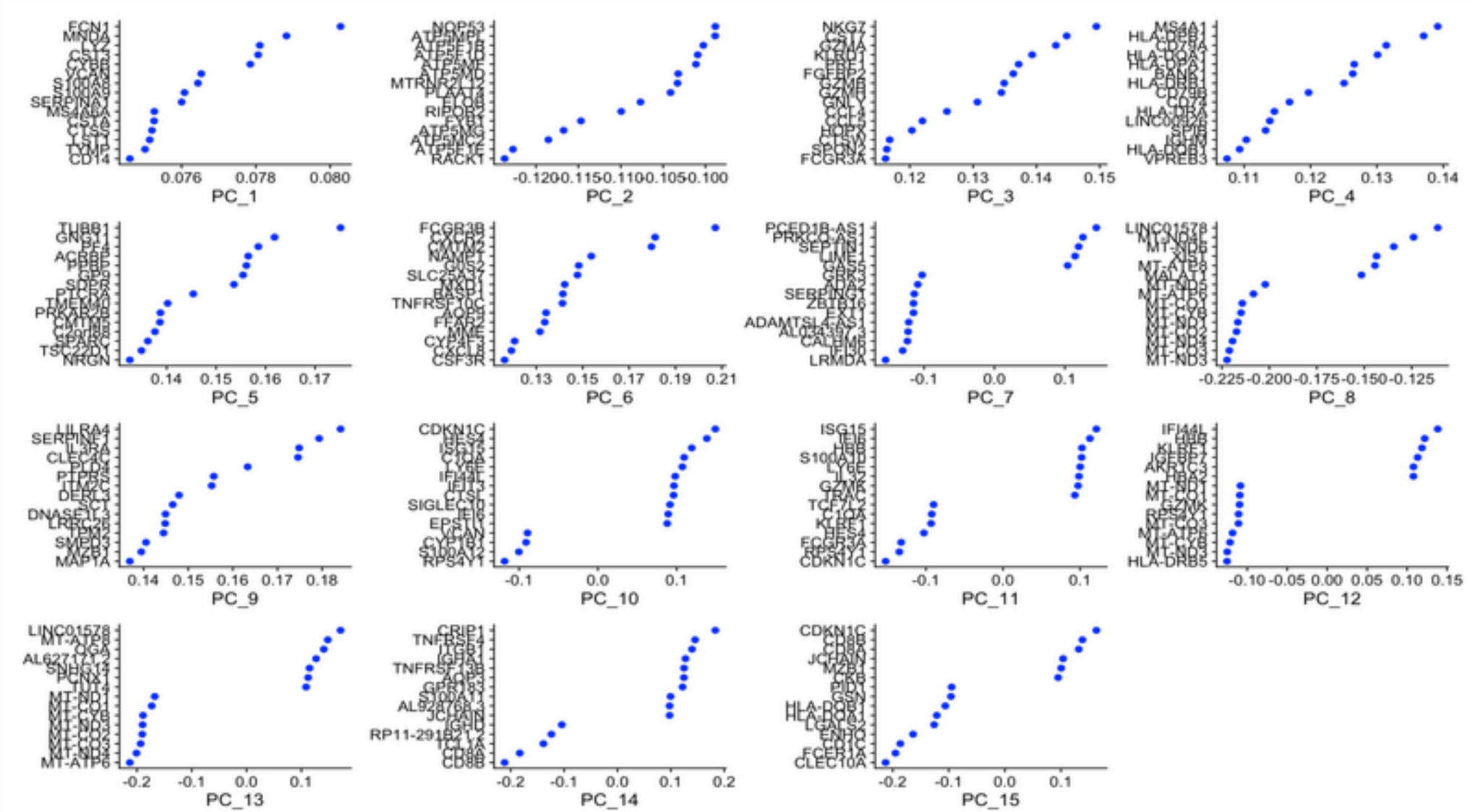
These PCs capture the most significant biological signals and major differences between cell types

After the Elbow:

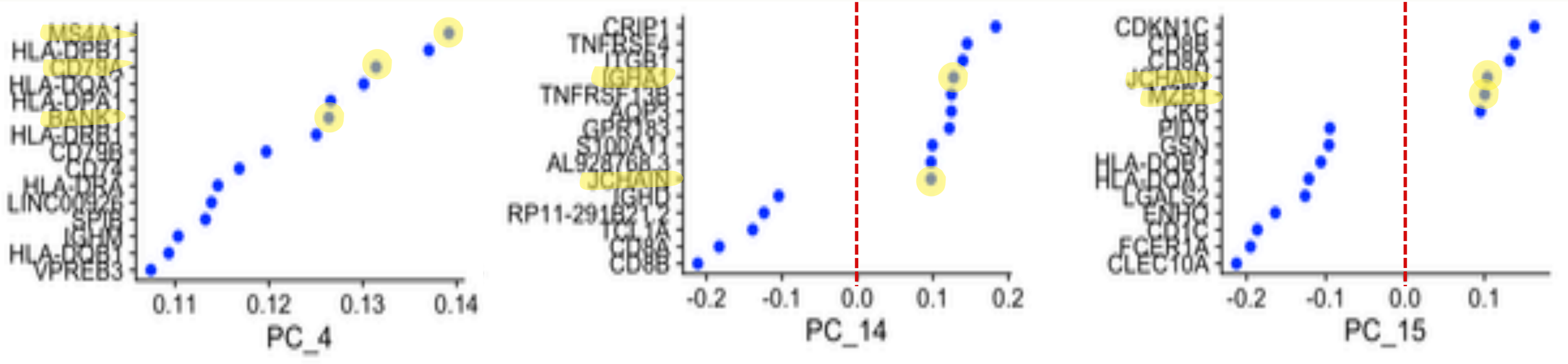
Typically represent "noise" or technical variation that does not contribute meaningfully to the biological clusters

Number of PCs to use for subsequent UMAP and clustering

Principle Component Analysis (PCA)



Principle Component Analysis (PCA)



B cell Markers

- **MS4A1**
- **CD79A**
- **BANK1**

These genes are expressed by cells that have differentiated into plasmablasts or plasma cells to produce autoantibodies:

- **JCHAIN**
- **MZB1**
- **IGHA1**

Harmony and UMAP

```
library(harmony)
merge <- harmony::RunHarmony(merge,
                             group.by.vars = 'Sample',
                             reduction.save = 'harmony')
table(merge@meta.data$Condition)
```

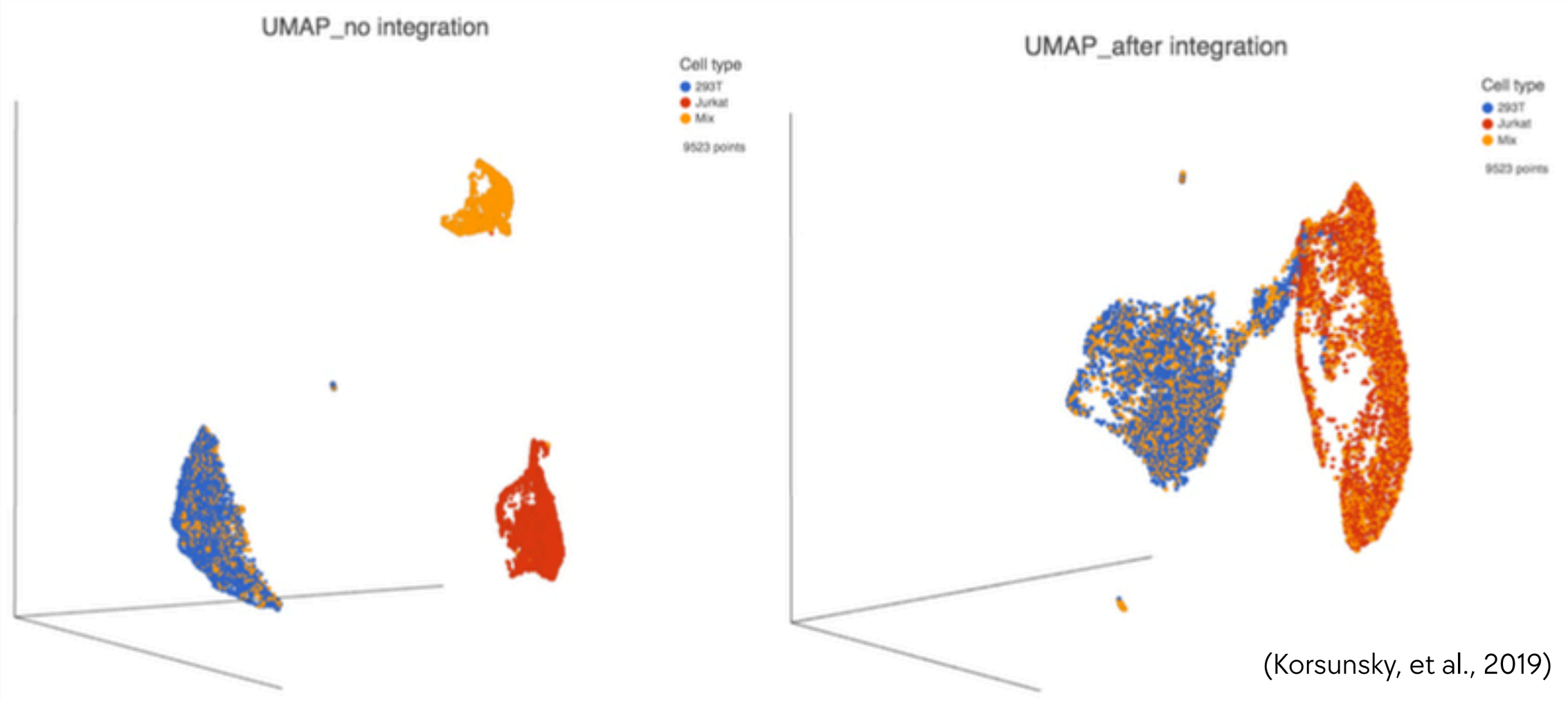
```
Harmony converged after 6 iterations
```

```
   HC    LN   SLE
36711 32999  6706
```

```
#visualize with umap after harmony

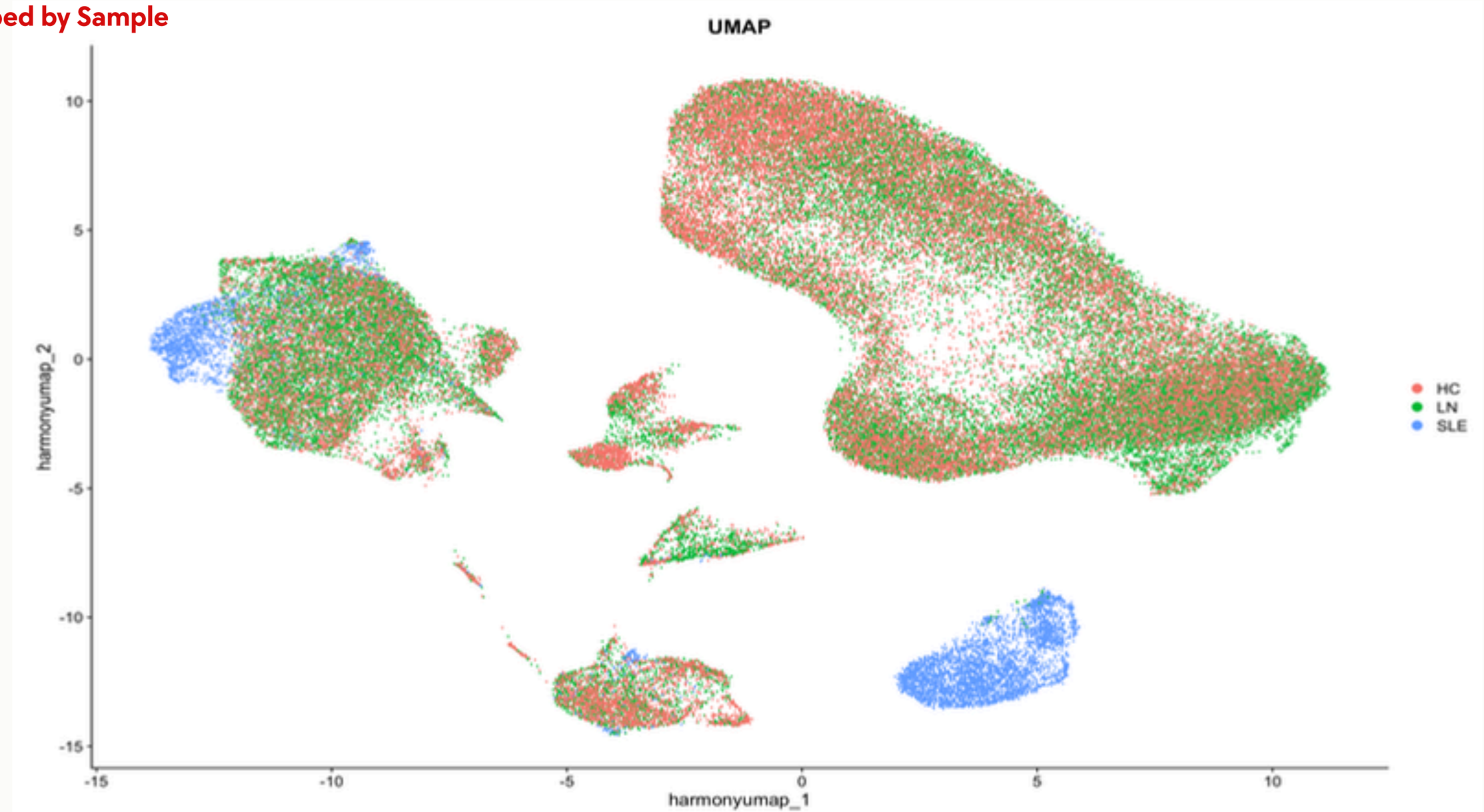
```{r, warning=FALSE, message=FALSE}
merge <- RunUMAP(merge,
                 dims = 1:8,
                 reduction = "harmony",
                 reduction.name = 'harmonyumap')
DimPlot(merge, group.by = 'Condition',reduction = 'harmonyumap') + labs(title = "UMAP")
```
```

Harmony and UMAP



(Korsunsky, et al., 2019)

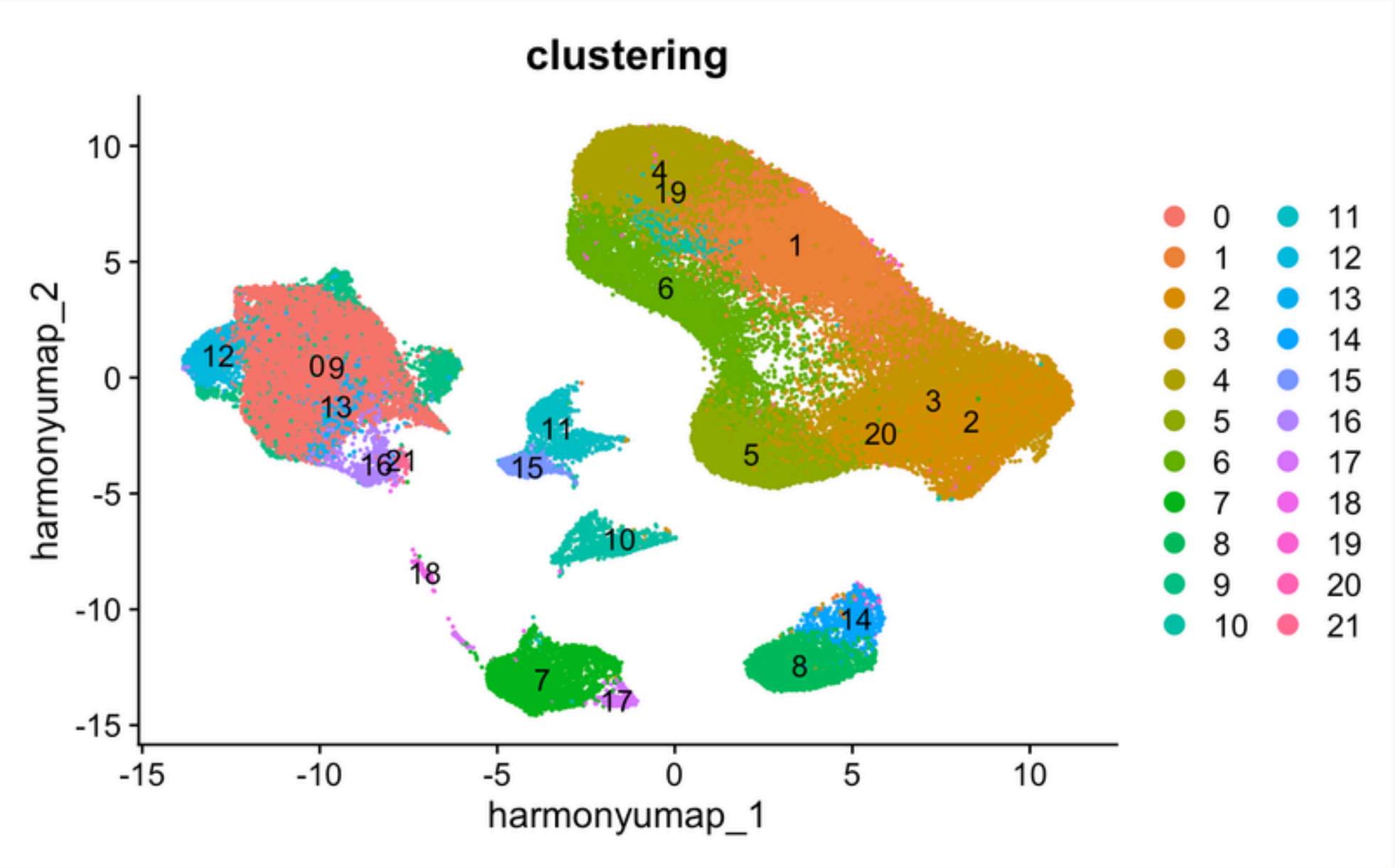
Grouped by Sample



Clustering: The Louvain Algorithm

```
### cluster analysis
#clustering separates the cells, Louvains Method

```{r, warning=FALSE}
merge <- FindNeighbors(merge, reduction = "harmony", dims = 1:15)
merge <- FindClusters(merge, resolution = 0.1)
merge <- FindClusters(merge, resolution = 0.5)
#resolution directly proportional with cluster, higher resolution, more clusters
DimPlot(merge, group.by = 'RNA_snn_res.0.1', reduction = 'harmonyumap', label = TRUE) + labs(title = "clustering")
DimPlot(merge, group.by = 'RNA_snn_res.0.5', reduction = 'harmonyumap', label = TRUE) + labs(title = "clustering")
```
```


clustering
harmonyumap_2
harmonyumap_1
10
5
0
-5
-10
-15
-15
-10
-5
0
5
10
4
19
1
6
12
0
9
13
11
15
16
21
3
2
20
5
10
18
14
8
7
17
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21

Annotation: Single R

```r
ref <- celldex::DatabaseImmuneCellExpressionData()

sce <- as.SingleCellExperiment(merge)

results <- SingleR(test = as.SingleCellExperiment(merge),
                   ref = ref, labels = ref$label.main)

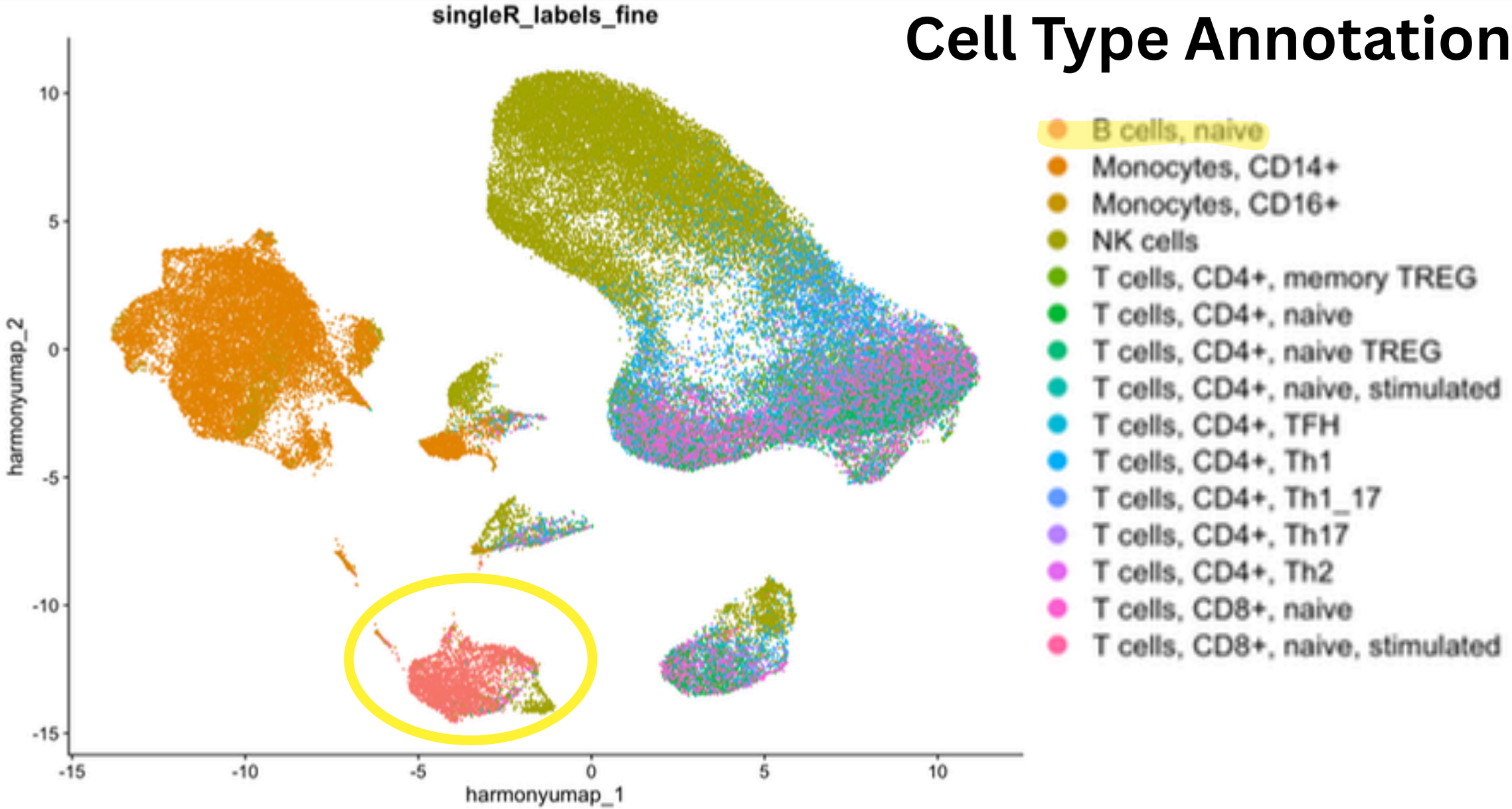
results_fine <- SingleR(test = as.SingleCellExperiment(merge),
                        ref = ref ,labels = ref$label.fine)

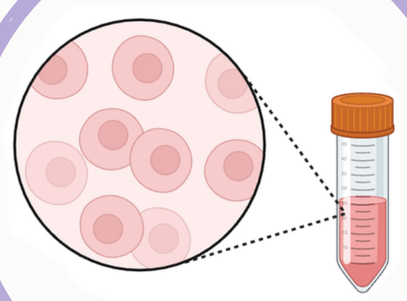
merge$singleR_labels_m <- results$labels
merge$singleR_labels_fine <- results_fine$labels

DimPlot(merge, reduction = 'harmonyumap', group.by = 'singleR_labels_m')
DimPlot(merge, reduction = 'harmonyumap', group.by = 'singleR_labels_fine')

table(merge@meta.data$singleR_labels_m)
table(merge@meta.data$singleR_labels_fine)
```

Cell Type Annotation





PBMC samples
(3 HC, 3 SLE , 3 LN)

Illumina NovaSeq 6000

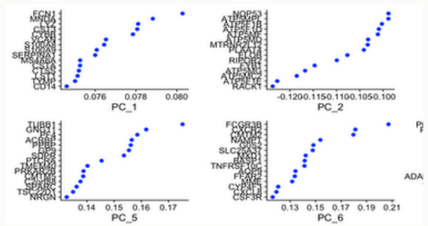


Single Cell RNA Sequencing



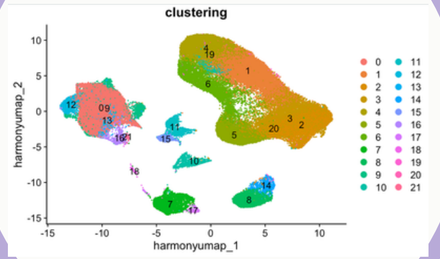
Quality Control, Normalization & Harmony Batch Correction

PCA



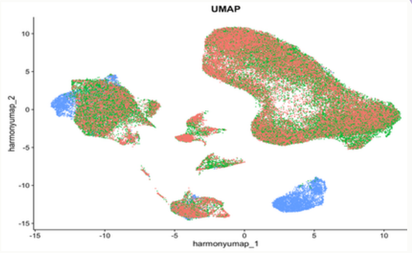
Reduce Data Dimensionality

Cell Custering



Identify Groups of Transcriptionally Similar Cells

UMAP

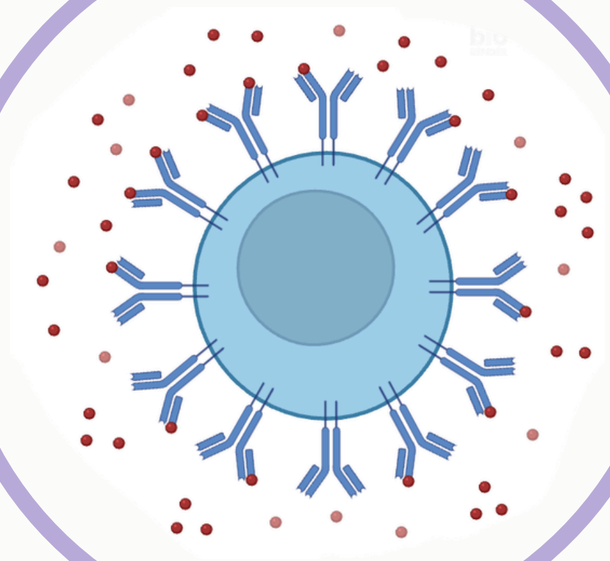


2D Relationship Visualization

- B cells, naive
- Monocytes, CD14+
- Monocytes, CD16+
- NK cells
- T cells, CD4+, memory TREG
- T cells, CD4+, naive
- T cells, CD4+, naive TREG
- T cells, CD4+, naive, stimulated
- T cells, CD4+, TFH
- T cells, CD4+, Th1
- T cells, CD4+, Th1_17
- T cells, CD4+, Th17
- T cells, CD4+, Th2
- T cells, CD8+, naive
- T cells, CD8+, naive, stimulated

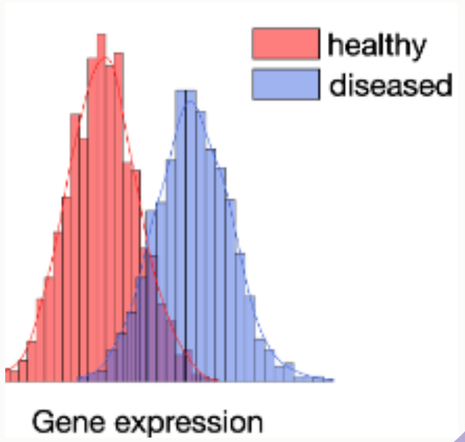
Cell Type Annotation

Future Direction

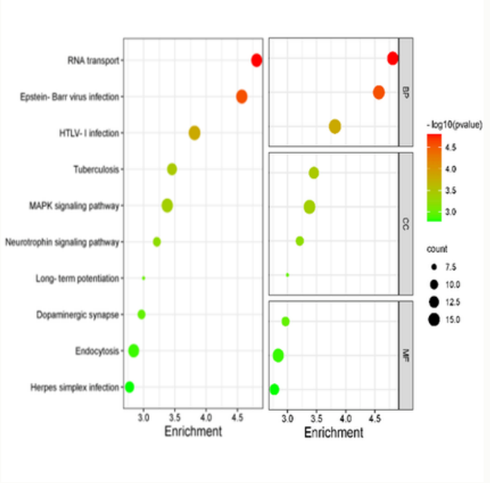


B-Cell Subsetting

DEG



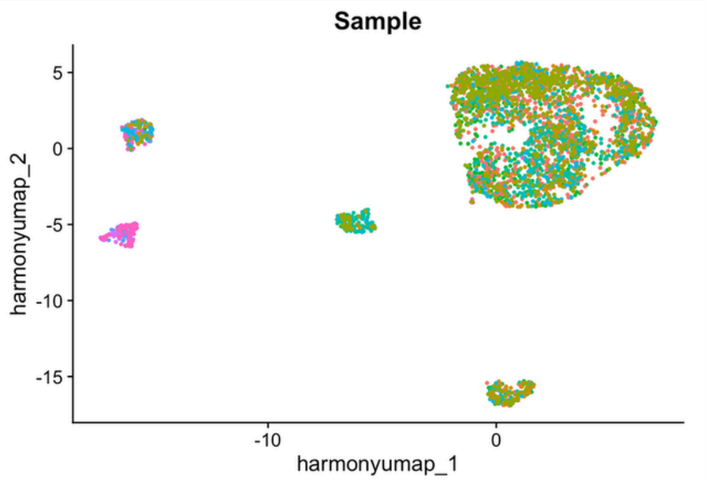
Identify abnormal genes and pathways



Pathway Enrichment Analysis

Future Direction

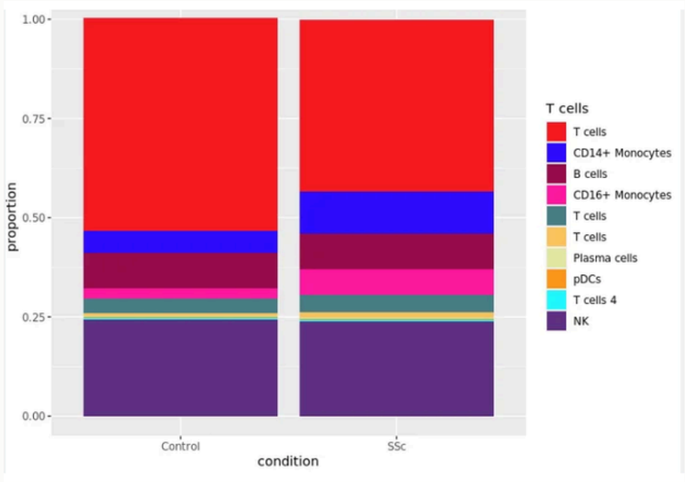
B-Cell Subset UMAP



Marker Heatmap

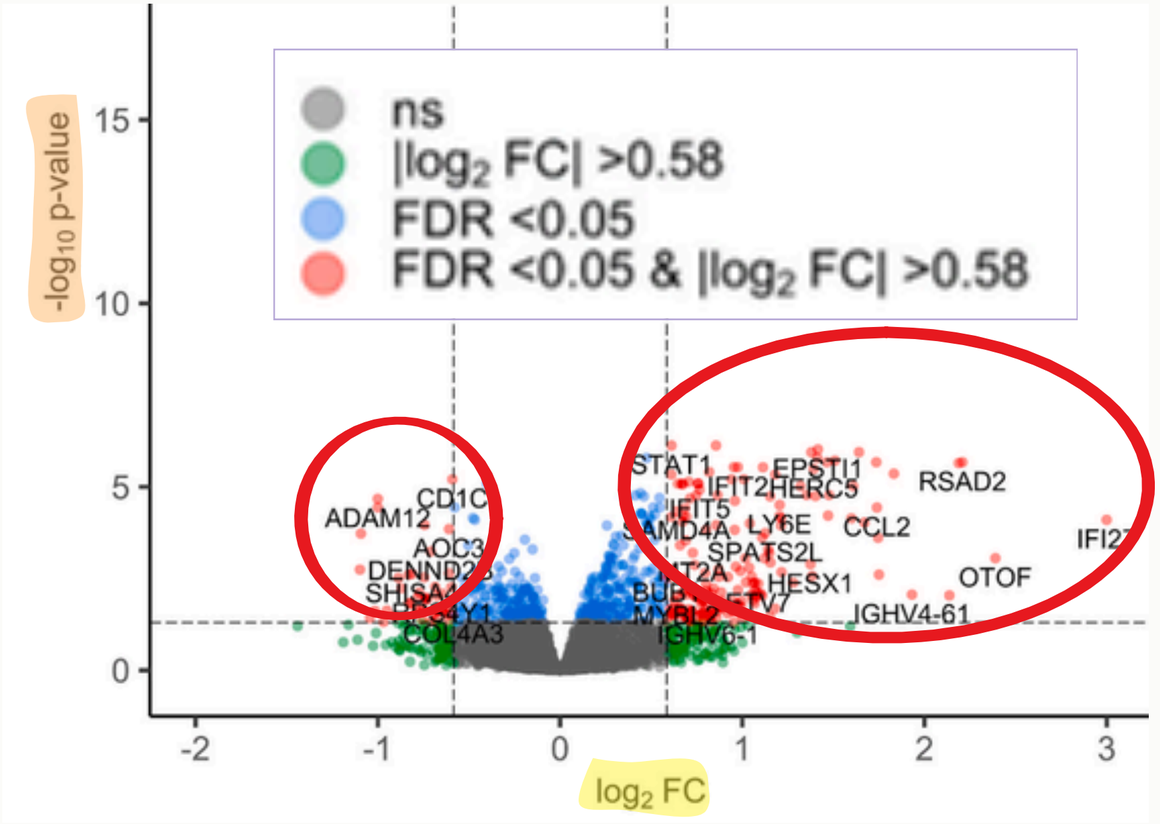


Composition Chart



Differential gene expression (DEG)

In any disease, some genes are overactive while others are suppressed. DEG compares RNA transcript levels to find significant differences.



(Figure from Lindbloom, et al., 2023)

Which genes are abnormal?

Each dot = 1 gene

Fold Change (log2 FC)	False Discovery Rate (FDR)
Magnitude Is the gene upregulated or downregulated? Study Threshold: >|0.58|	Confidence/Statistical Significance Corrects for random chance Ensures that biological signal is real Study Threshold: <0.05

Results:

521 genes successfully passed both filters

Significance

- Improved understanding of B-cell transcriptional changes in SLE and LN
- Identification of candidate biomarkers for disease progression
- Logical workflow/baseline results for future verification studies
- Potential therapeutic targets for precision medicine

Future Direction

- Continue this ongoing study

- Validate findings in larger independent cohorts
- Functionally characterize candidate genes and pathways
- Investigate B-cell subtype-specific transcriptional changes
- Integrate multi-omics and clinical data
- Explore biomarkers for prognosis and therapeutic response

Acknowledgements





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